

Notices & Litigation

Phase 3 — Portal Autopilot

Milestone M4 · "fresh data every morning for ≤ 20 minutes of human time" · 6 October 2026 · branch `claude/kind-cerf-jbqu8m`, PR #116

15 / 15

SQL test suites pass, each migration applied twice (new: `test_99_autopilot`)

7 / 7

agent end-to-end tests: the real agent and the real extension against a stand-in portal

44 / 44

extension simulation checks, the 0.6.0 scenarios included

28 / 28

app checks on the Autopilot page and CAPTCHA wall; every screen at 1440 and 390 px with 0 accessibility violations

Ships switched off. Nothing reaches the GST portal until a manager switches the autopilot on (Notices → Autopilot → Settings), and the two e-mails it can send (E16 nudge, E17 "ask the client") are off like every alert. **To go live (§6):** run the four migrations (copy page), merge PR #116, set up the office PC (one installer), reload the extension 0.6.0 on staff PCs, and the partner's sign-off on the decisions in §5.

Contents

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2. How it works, and a morning with it
3. Screens
4. How it was verified, and what could not be
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7. Acceptance criteria — how they are measured; what is next

1. What shipped, by roadmap task

Task 1 · The agent's notices bundle Done

The old agent (agent/) is rewritten. Each of up to four "workers" is a Chromium profile with **the same GST Keeper extension the staff use**, copied from the repository at every start. A worker claims a job (`portal_job_claim`: one atomic claim with SKIP LOCKED, highest priority first), starts it in the extension (`startAgentJob`), relays the login CAPTCHA to the app, and finishes the job from what the extension wrote to the run ledger. Logging in, reading notices, case folders, refunds, DRC-03, applications and the profile, and saving them through `sync_ingest` is the extension's own code, unchanged from a person's Sync.

- **Retries with backoff**: a portal error or time-out goes back on the queue after 5, 10, 20 minutes (3 attempts, setting); a rejected password does not retry.
- **Sessions**: by default the agent logs out after each client, like a person. With "Keep portal sessions" on, a client's portal cookies are kept **encrypted on the office PC only** (AES-256-GCM) for up to 50 minutes so a second job needs no CAPTCHA. Nothing is stored in the database.
- **No secret on the PC**: the agent uses the app's publishable key. The old agent needed the service-role key.
- Deleted: the CAPTCHA-solver "seam", the unfinished save-to-portal handlers, and the cloud runners (GitHub Actions workflow, Render, Docker).

Task 2 · Schedule and CAPTCHA wall Done

- **Schedule**: a `pg_cron` clock every 5 minutes (`autopilot_tick`) queues every active client at **05:30 IST** and the urgent ones at **13:00** (notices due within 7 days or overdue, never synced, stale for 20 h, or failing), sends the **09:00 nudge** and at **23:30** closes the day: whatever still waits gets a named reason (`captcha_timeout`, `not_reached`, `agent_offline`).
- **CAPTCHA wall** (`/notices-autopilot`): the agent opens a portal login only while somebody is attentive at the wall — the wall on screen, or typed on in the last 10 minutes. Otherwise the job is parked as "waiting for a CAPTCHA" **without the portal being opened at all**. Type, Enter, next; "Can't read it" fetches a new one; "Skip this client" leaves it for today. A header badge shows how many are waiting, everywhere in the app.
- **Failure reasons with one-click fixes** ("Needs a person"): password rejected → update it / ask the client; account locked → ask the client to reset (OTP); CAPTCHA never typed → open the wall; registration cancelled → mark the client inactive; portal errors → run again.

Task 3 · More read in the same session Partly

- **Applications on the portal**: appeals (APL-01), rectification, objections to a provisional attachment, the s.128A waiver, compounding and provisional assessment — the portal's own "My Applications" codes, taken from its public `caserearchctrl.js`, not guessed. Shown on the client's page.
- **Registration status** (Active / Cancelled / Suspended) from the taxpayer profile in every sync.
- **GSTR-3A**: the return type and period are kept (the portal's own `gstr3actrl.js` confirms `MMYYYY`) and the notice closes itself once Filing Status shows that return filed; the rebuilt PDF now says it is rebuilt (S-31).
- **Not built yet**: Liability Register Part II, Rule 86A blocked credit, DRC-01B/01C Part-B status, bulk filing status, and refund folders inside the daily bundle (refund documents remain their own pull, now queueable from "Fetch a report"). Their endpoints are to be confirmed from the portal's public scripts or a recorded staff session first.

Task 4 · Portal e-mails trigger syncs Done

The agent reads the firm's notices inbox (IMAP, read-only — nothing deleted, moved or marked), picks the GSTIN (check digit validated), form and reference out of each portal e-mail, direct or forwarded, and queues a **priority sync** for that client. Routine portal e-mails (OTPs, filing acknowledgements, receipts) queue nothing. Only the subject, a 400-character snippet (OTPs masked) and the extracted fields are stored. When the notice arrives, the e-mail is linked to it — that is how "within 4 hours" is measured.

Task 5 · Fetch a report now Done

The Autopilot page queues any of the extension's pulls (filed GSTR-1 / 3B / 9, 2A, 2B, ledgers, refunds, DRC-03, challans, profile...) for chosen clients and periods; results land in the same reports as an extension pull. "Sync now" (and the extension's popup) can send its clients to the office agent instead of running in the person's own browser — no extension needed on that PC.

Task 6 · Written CAPTCHA posture Done — for sign-off

`docs/PORTAL_AUTOPILOT_POSITIONS.md`: people type every CAPTCHA; no solver, OCR or solving service is built or used; the agent runs only in the office, on the office network, and never disguises its browser (no proxy, no IP rotation, no user-agent or `navigator.webdriver` masking); client authorisations on file. See §5.

2. How it works, and a morning with it

Queue 05:30 all active clients · 13:00 the urgent ones · a portal e-mail · "Fetch a report"	Claim The office agent takes the highest-priority job; nobody at the wall → it waits, portal untouched	CAPTCHA The login page's CAPTCHA appears on the app's wall; a person types it	Read and save The extension logs in, reads notices, cases, refunds, DRC-03, applications, profile; saves through sync_ingest	Finish Log out; job done (or retried, or a named reason); the run ledger and the Autopilot page show it
05:30	The clock queues ~95 clients. Nobody is at the wall, so they wait as "waiting for a CAPTCHA"; the portal is not opened.			
09:00	The header badge says "95 waiting for a CAPTCHA" (and, once switched on, an e-mail nudge goes to the team).			
09:05	One person opens Notices → Autopilot → CAPTCHA wall. Within seconds the first CAPTCHAs appear (one per browser). Type, Enter: the agent logs that client in and reads it while the next CAPTCHA is typed. The person can work in another tab — the tab title and an optional desktop notice say when a CAPTCHA is waiting.			
~10:30	All clients read (≈ 1–2 minutes of portal work per client per browser). About 95 CAPTCHAs at ~8 seconds each: about 13 minutes of human time , measured per CAPTCHA by the wall.			
13:00	The urgent clients again (due within 7 days, failing, stale).			
23:30	The day closes: every client is fresh or has a named reason — the acceptance measure.			

Why the wall and not a solver. The portal's CAPTCHA is its way of knowing a person is logging in. The autopilot keeps that true: a person types every one, and the agent only ever opens a login when a person is there to type it. What the autopilot removes is the waiting — the hours of watching a browser go client by client — not the person.

What changed in the data model

Object	What it is
autopilot_settings	Master switch (off), pause, schedule times and scope, browsers at once, keep sessions, e-mail trigger, retries, CAPTCHA refresh.
portal_jobs	Gains priority, origin, run, backoff, reason class, CAPTCHA timings (shown, answered, time at the CAPTCHA, by whom), session reused; new status waiting_captcha.
autopilot_presence · autopilot_wall_minutes	Who is attentive at the wall now; attentive minutes per person per day.
autopilot_slot_runs	Each schedule slot fires once per IST day (morning, afternoon, nudge, close).
portal_emails	Portal e-mails read from the inbox: extracted fields, client, status, the job, the notice once it arrives.
gst_portal_applications	Applications on the portal per client (appeals and others).
gst_notices.portal_detail	Detail beyond the list row (GSTR-3A return and period).
RPCs	autopilot_enqueue, portal_job_claim / start / park / captcha / answer / finish / cancel / retry, autopilot_wall_ping, autopilot_badge, autopilot_heartbeat, autopilot_status, autopilot_metrics, autopilot_ask_client, portal_email_ingest, sync_ingest_applications, sync_notice_details, notices_close_gstr3a_filed.

3. Screens

Local build against demo data (the autopilot switched on, an agent online, five clients waiting); desktop 1440 px (cropped to the top) and phone 390 px.

Notices & Litigation - Portal autopilot

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · View all CAPTCHA today 0.5 min · wall open (pending) 18.4 min

Command centre · Work queue · All notices · Matters · Hearings · Calendar · Clients · Autopilot · Reports

CAPTCHA wall · Queue · Needs a person · Fetch a report · Portal e-mails · Acceptance · Settings

CAPTCHA wall

5 clients waiting for a CAPTCHA · 3 on screen. Type each one and press Enter: the next box takes the cursor while the agent logs that client in.

Omega Retail · Gamma Test Exports · Demo Textiles Pvt Ltd

730264 · 159837 · 482915

Type the digits you see · Can't read it · Skip this client

2 more clients waiting: the agent opens the next logos on its type.

On the office PC: Demo Textiles Pvt Ltd waiting for a CAPTCHA · Sample Pharma LLP, reading the portal.

CAPTCHA images expire after a few minutes, so the agent replaces one nobody typed within 150 seconds. Nobody at the wall means nothing is opened: the agent parks those clients as waiting and brings them back the ...

CAPTCHA wall — type, Enter, the next box takes the cursor

Notices & Litigation - Portal autopilot

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · View all CAPTCHA today 0.5 min · wall open (pending) 18.4 min

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CAPTCHA wall · Queue · Needs a person · Fetch a report · Portal e-mails · Acceptance · Settings

No portal password saved · 1 client

No portal password is saved for these clients. Add it, or ask the client for it, the next run logs in with it.

Iota Demo Industries · Delta Demo Traders

Update password · Ask the client

Portal rejected the saved password · 1 client

The portal refused the saved password, so it was probably changed. Update it, or ask the client for the new one, then bring them back the portal account, so to fix it before running the client again.

Delta Demo Traders

Update password · Ask the client

Portal account locked · 1 client

The portal account is locked. Only the client can reset it (Forgot Password: the OTP goes to their registered mobile and e-mail). Ask them, then save the new password.

Test Engineering Works

Ask the client to reset (OTP) · Save the new password

Nobody typed the CAPTCHA · 1 client

Nobody typed the CAPTCHA for these clients. Run them again and type the CAPTCHA on the wall.

Sigma Logistics

Open the CAPTCHA wall · Run again

Skipped on the CAPTCHA wall · 1 client

Scheduled skipped these clients on the wall, then bring them again when there is time to type their CAPTCHA.

Lambda Sample Exports

Open the CAPTCHA wall · Run again

Portal error · 1 client

Usually temporary. Run them again; if one keeps failing, open its sign log.

Kappa Chemicals

Run again

Registration cancelled or suspended · 2 clients

The portal's company profile shows these active clients' registration as cancelled or suspended. Check with the client; if the firm no longer acts, mark the client inactive so the schedule stops logging in.

Kappa Chemicals · Sigma Logistics

Mark inactive · Mark inactive

Needs a person — each failure grouped by its fix

Notices & Litigation - Portal autopilot

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · View all CAPTCHA today 0.5 min · wall open (pending) 18.4 min

Command centre · Work queue · All notices · Matters · Hearings · Calendar · Clients · Autopilot · Reports

CAPTCHA wall · Queue · Needs a person · Fetch a report · Portal e-mails · Acceptance · Settings

Client · What · Run · Status · Times · When · CAPTCHA's

Omega Retail · Gamma Test Exports · Demo Textiles Pvt Ltd · Beta Sample Intra LLP · Sample Pharma LLP · Iota Demo Industries · Kappa Chemicals · Omega Retail · Demo Textiles Pvt Ltd · Sigma Logistics · Delta Demo Traders · Lambda Sample Exports · Test Engineering Works · Sample Pharma LLP · Gamma Test Exports

What · Run · Status · Times · When · CAPTCHA's

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Queue — today's jobs, reasons, tries, who typed the CAPTCHA

Notices & Litigation - Command centre

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · View all CAPTCHA today 0.5 min · wall open (pending) 18.4 min

Command centre · Work queue · All notices · Matters · Hearings · Calendar · Clients · Autopilot · Reports

CAPTCHA wall · Queue · Needs a person · Fetch a report · Portal e-mails · Acceptance · Settings

Overview & still open · 17 · Done in last 7 days · 4 · New in last 24 h · Without an owner · 0 · In partner · 2

Today's plan — next best action

Client · Notice · what the system read · Due · Readiness · Next action · Owl

Beta Sample Intra LLP · Notice for personal hearing - FY 2023-24 · 15 d later · not started · Write draft · 1

REG-17 · Show cause notice for cancellation of registration - FY 2022-23 · 15 d later · 0 of 2 documents in · Chase client (2) · 3

Beta Sample Intra LLP · DRC-07 · Summary of the demand order - FY 2023-24 · 15 d later · not read yet · Triage · 3

Kappa Chemicals · GSTR-3A · Notice to a return defaulter (GSTR-3A, s. 44B) - FY 2022-23 · 15 d later · not read yet · Triage · 3

Test Engineering Works · DRC-01A · Intimation of tax ascertained before a show cause notice - FY 2023-23 · 15 d later · not read yet · Triage · 3

Kappa Chemicals · DRC-01A · Intimation of tax ascertained before a show cause notice - FY 2023-23 · 15 d later · not read yet · Triage · 3

Delta Demo Traders · Notice for personal hearing - FY 2023-22 · 15 d later · not read yet · Triage · 3

Sync now — "Send to the office agent", no extension needed

Notices & Litigation - Portal autopilot

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · View all CAPTCHA today 0.5 min · wall open (pending) 18.4 min

Command centre · Work queue · All notices · Matters · Hearings · Calendar · Clients · Autopilot · Reports

CAPTCHA wall · Queue · Needs a person · Fetch a report · Portal e-mails · Acceptance · Settings

Fetch a report for any client

Client · What · Run · Status · Times · When · CAPTCHA's

Omega Retail · Gamma Test Exports · Demo Textiles Pvt Ltd · Beta Sample Intra LLP · Sample Pharma LLP · Iota Demo Industries · Kappa Chemicals · Omega Retail · Demo Textiles Pvt Ltd · Sigma Logistics · Delta Demo Traders · Lambda Sample Exports · Test Engineering Works · Sample Pharma LLP · Gamma Test Exports

What · Run · Status · Times · When · CAPTCHA's

Omega Retail · Gamma Test Exports · Demo Textiles Pvt Ltd · Beta Sample Intra LLP · Sample Pharma LLP · Iota Demo Industries · Kappa Chemicals · Omega Retail · Demo Textiles Pvt Ltd · Sigma Logistics · Delta Demo Traders · Lambda Sample Exports · Test Engineering Works · Sample Pharma LLP · Gamma Test Exports

Queue — today's jobs, reasons, tries, who typed the CAPTCHA

Notices & Litigation - Portal autopilot

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · View all CAPTCHA today 0.5 min · wall open (pending) 18.4 min

Command centre · Work queue · All notices · Matters · Hearings · Calendar · Clients · Autopilot · Reports

CAPTCHA wall · Queue · Needs a person · Fetch a report · Portal e-mails · Acceptance · Settings

Where the results land

Client · What · Run · Status · Times · When · CAPTCHA's

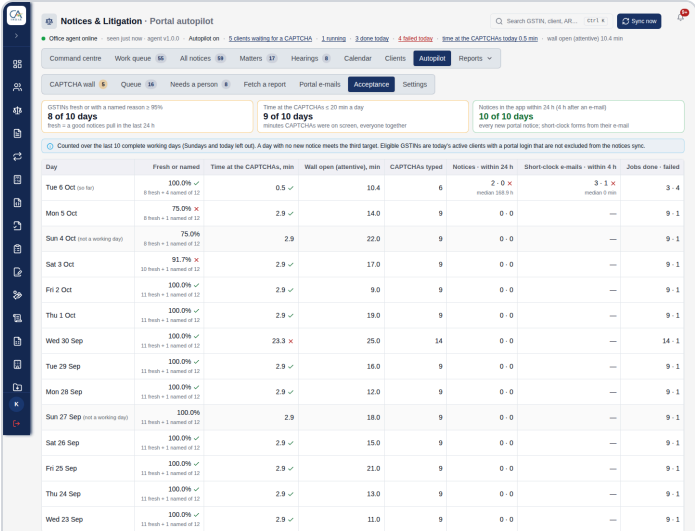
Omega Retail · Gamma Test Exports · Demo Textiles Pvt Ltd · Beta Sample Intra LLP · Sample Pharma LLP · Iota Demo Industries · Kappa Chemicals · Omega Retail · Demo Textiles Pvt Ltd · Sigma Logistics · Delta Demo Traders · Lambda Sample Exports · Test Engineering Works · Sample Pharma LLP · Gamma Test Exports

What · Run · Status · Times · When · CAPTCHA's

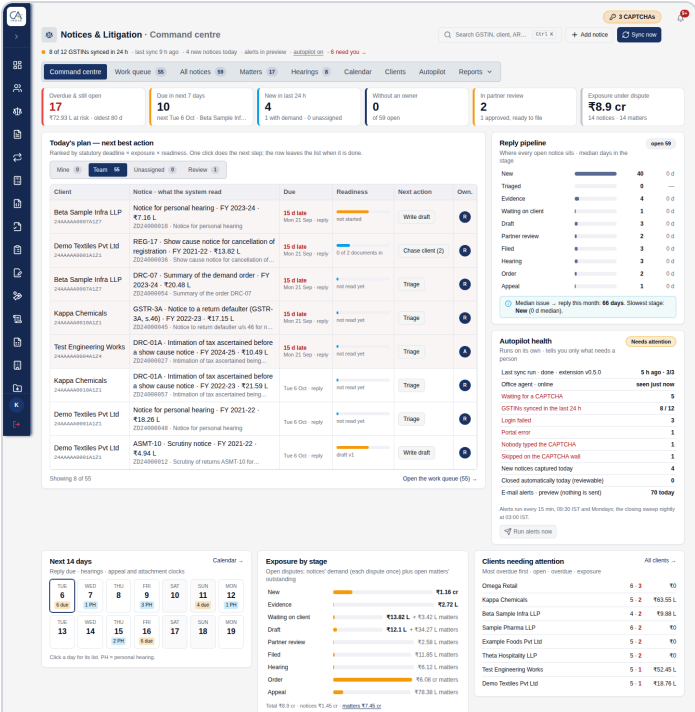
Omega Retail · Gamma Test Exports · Demo Textiles Pvt Ltd · Beta Sample Intra LLP · Sample Pharma LLP · Iota Demo Industries · Kappa Chemicals · Omega Retail · Demo Textiles Pvt Ltd · Sigma Logistics · Delta Demo Traders · Lambda Sample Exports · Test Engineering Works · Sample Pharma LLP · Gamma Test Exports

Queue — today's jobs, reasons, tries, who typed the CAPTCHA

Fetch a report — any extension pull, chosen clients and periods

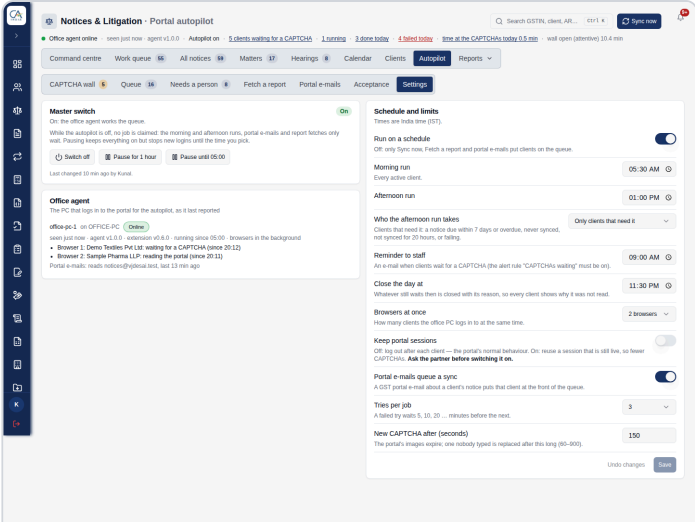


Acceptance — the three targets over 10 working days

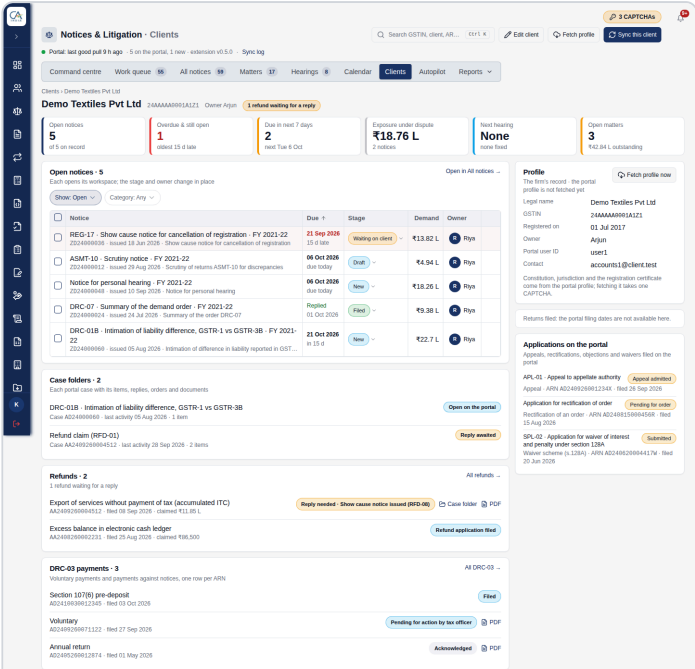


Command centre — header badge, office agent and CAPTCHA rows

Portal e-mails — matched client, queued sync, e-mail to app time



Settings — master switch, pause, schedule, browsers (managers)



Client page — applications on the portal

V J Desai & Co. LLP

CHARTERED ACCOUNTANTS

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in.

6 typed today · 3 done today · Also at the wall: Priya

You can work in another tab: the agent keeps fetching CAPTCHAs while you type one at least every 10 minutes, and this tab's title shows how many are waiting.

Omega Retail

24AAAAA0905A1Z8

Portal e-mail

Try 2

The portal said the last one was wrong; this is a new CAPTCHA.

Type the digits you see

Send

Can't read it

Skip this client

Gamma Test Exports

24AAAAA0905A1Z5

GSTR-3B as filed

Type the digits you see

Send

Can't read it

Skip this client

Demo Textiles Pvt Ltd

24AAAAA0901A1Z1

Morning run

Phone · CAPTCHA wall

V J Desai & Co. LLP

CHARTERED ACCOUNTANTS

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Notices & Litigation · Portal autopilot

Search GSTIN, client, ARN, DIN...

Sync now

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · time at the CAPTCHAs today 0.5 min · wall open (attentive) 10.7 min

Autopilot

CAPTCHA wall 5 Queue 16

Needs a person 8 Fetch a report

Portal e-mails Acceptance Settings

No portal password saved 1 client

No portal password is saved for these clients. Add it, or ask the client for it; the next run logs in with it.

Iota Demo Industries 24AAAAA0913A1Z4

"No saved GST portal password for this client."

Failed 8 h ago · last good pull never · sync log

Update password Ask the client

Portal rejected the saved password 1 client

The portal refused the saved password, so it was probably changed. Update it, or ask the client for the new one; three wrong tries lock the portal account, so fix it before running the client again.

Delta Demo Traders 24AAAAA0906A1Z6

Phone · needs a person

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CHARTERED ACCOUNTANTS

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Notices & Litigation · Portal autopilot

Search GSTIN, client, ARN, DIN...

Sync now

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · time at the CAPTCHAs today 0.5 min · wall open (attentive) 10.7 min

Autopilot

CAPTCHA wall 5 Queue 16

Needs a person 8 Fetch a report

Portal e-mails Acceptance Settings

Client or GSTIN

Status: Every job today (16) Run: Any Refresh

Omega Retail 24AAAAA0905A1Z8

CAPTCHA on the wall

Notices, refunds and DRC-03

Portal e-mail - started 20:09 · 2 CAPTCHAs · typed by Priya

Cancel

Gamma Test Exports 24AAAAA0905A1Z5

GSTR-3B as filed · Jul – Sep 2026

Report fetch · Kunal · started 20:11 · 1 CAPTCHA

Phone · queue as cards

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CHARTERED ACCOUNTANTS

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Notices & Litigation · Portal autopilot

Search GSTIN, client, ARN, DIN...

Sync now

Office agent online · seen just now · agent v1.0.0 · Autopilot on · 5 clients waiting for a CAPTCHA · 1 running · 3 done today · 4 failed today · time at the CAPTCHAs today 0.5 min · wall open (attentive) 10.7 min

Autopilot

CAPTCHA wall 5 Queue 16

Needs a person 8 Fetch a report

Portal e-mails Acceptance Settings

Master switch On

On: the office agent works the queue.

While the autopilot is off, no job is claimed: the morning and afternoon runs, portal e-mails and report fetches only wait. Pausing keeps everything on but stops new logins until the time you pick.

Switch off Pause for 1 hour

Pause until 05:00

Last changed 10 min ago by Kunal.

Office agent

The PC that logs in to the portal for the autopilot, as it last reported

office-pc-1 on OFFICE-PC Online

Phone · settings

4. How it was verified, and what could not be

Check	What it proves	Result
SQL suites (local PostgreSQL, every migration applied twice)	Queue, dedupe, risk order, claim with SKIP LOCKED, park, CAPTCHA, stale answers refused, attentive presence, retries and backoff, named ledger reasons, run accounting, dead-agent release, the schedule firing once a day, portal e-mails (match, ignore routine, link), applications ingest (never removes on an empty type), GSTR-3A closing on filing, status and metrics, "ask the client" respecting the switches.	15 / 15
Extension simulation (real background.js, db.js, content.js in Node)	Phase 1–2 behaviour unchanged; 0.6.0: applications saved, registration status, GSTR-3A period kept and closed on filing, agent jobs left to the agent by the watchdog, startAgentJob / agentJobState / agentClearJob.	44 / 44
Agent end-to-end (real agent, real extension 0.6.0, Chromium, stand-in portal, local database)	Switched off → nothing touched; nobody at the wall → waits without opening the portal; wrong CAPTCHA then right → logged in, notices and applications saved, logged out; rejected password → failed with the reason, grouped as a password fix; skipped at the wall → named reason; kept session → second job with no CAPTCHA; heartbeat.	7 / 7
Inbox reader unit tests (synthetic e-mails, fake IMAP server)	Direct and forwarded portal e-mails (Gmail, Outlook, HTML-only, as attachment), GSTIN check digits, form and reference, routine e-mails, read-only IMAP, state and retries, slow-input guard.	52 / 52
Browser pass on the new pages (390 px and 1440 px)	The seven Autopilot tabs, the command centre, the client page, All notices and Clients at 1440 and 390 px: no sideways scroll and 0 accessibility violations (axe) on all 20 views; the switched-off, paused, agent-offline, idle and read-only (employee) states as well. 28 interaction checks: Enter sends and moves to the next CAPTCHA; the wall keeps pinging while hidden with the right "attentive" flag; the time on screen sent with each answer is within 0.1 s of the measured time; Can't read it; Skip; the desktop notice; Sync now sends to the agent without the extension; Retry and Cancel; Fetch a report over a month range; "Ask the client" says plainly it was not sent while client e-mails are off; Mark inactive; Pause, Resume and Save.	20 / 20 · 28 / 28
Live schema dry run	All four migrations applied to project gcquafqxbykxbexcdpy inside a transaction that was then rolled back: they apply cleanly on the live schema (95 eligible clients read by the status), and nothing remained afterwards.	Clean
App build and type gate	npm run build, npm run typecheck and the agent's typecheck on every commit.	Green

Not verifiable from here: the real GST portal. The agent has not logged into the live portal — that needs the office PC, saved credentials and the partner's go-ahead, and the portal refuses data-centre addresses. The login selectors are the ones confirmed on the live page in July; everything after login is the extension's existing, production code. If the portal refuses the agent's browser, the position is not to disguise it: staff keep using the extension (\$5).

5. Decisions for the partner

Decision	What is proposed	Default today
CAPTCHA posture	People type every CAPTCHA on the wall; no solver, OCR or service; the agent opens logins only when someone is at the wall.	Built this way
Client authorisations	Automated access with stored credentials is against GSTN's terms (2023 advisory) and carries IT Act exposure. Use the autopilot only for clients whose written authorisation is on file; mark the rest "Exclude from notices sync".	Partner to confirm
Agent host	An always-on PC on the office network (a mini-PC is enough), signed in, never sleeping; visible browser windows.	To be set up
Notices inbox	A notices@ mailbox the agent reads read-only; clients forward portal e-mails to it, or the firm is the registered contact. Google Workspace with an app password works; Microsoft 365 needs a forward (no IMAP passwords).	Off (not configured)
Keep portal sessions	Off: log out after each client. On: fewer CAPTCHAs, sessions kept encrypted on the PC for 50 minutes.	Off
Referer rewrite (S-29)	The extension rewrites the Referer of one portal call to rebuild GSTR-3A PDFs (since September). Keep, or drop and keep GSTR-3A notices without a rebuilt PDF.	Unchanged (kept)
E-mails E16 and E17	The 09:00 nudge to the team; "ask the client" for a rejected password or a locked account (asks for the password by a secure channel, not e-mail).	Off
Afternoon run	13:00 urgent clients only ("priority"), every client ("all"), or none.	Priority

6. Going live, step by step

1. **Database:** run the four migrations from the copy page (Supabase SQL editor, in order; each says "Success. No rows returned"; each has a check). The autopilot stays off.
2. **App:** merge PR #116; `gst.vjdesai.com` deploys with the Autopilot page (it shows "agent offline" until step 3).
3. **Office PC:** copy the repository, double-click `agent/install-agent.bat` (Node.js 20+ first). It asks for nothing secret. Set the PC never to sleep. The Autopilot page then shows the agent online.
4. **Staff PCs:** reload the extension (0.6.0) in Chrome — the app recommends it.
5. **Try one client:** keep the autopilot off, open the wall, and from "Fetch a report" queue one client; switch the autopilot on, type the CAPTCHA, watch it finish in the Queue tab; switch off again if anything looks wrong.
6. **Switch on** (Settings) with 2 browsers and the schedule on; type CAPTCHAs each morning; watch the Acceptance tab for 10 working days.
7. **Later, one by one:** the inbox (`agent/.env`), keep sessions, E16, E17.

Rollback is the switch. Turning the autopilot off stops every new portal login at once (a client already logged in finishes its pull). Nothing else depends on it: staff can always sync with the extension as before.

7. Acceptance criteria — how they are measured; what is next

Roadmap criterion	How the app measures it (Acceptance tab, <code>autopilot_metrics</code>)
For 10 consecutive working days $\geq 95\%$ of active GSTINS are fresh (< 24 h) or carry a named failure reason	Per IST day: clients with portal credentials, not inactive or excluded; fresh = a notices pull succeeded in the 24 h before the day's end; named = not fresh but a login or notices attempt in those 24 h failed with a reason. Working days Monday–Saturday.
Human time on fetching ≤ 20 minutes a day (from about 3 hours)	Time each CAPTCHA was on screen in front of a person until they submitted it (the wall's clock pauses while the tab is hidden), summed per day; attentive wall minutes shown alongside.
A notice on the portal is in the app within 24 h, and within 4 h for 7-day forms	New portal notices (issued at most 10 days before capture) captured before the end of the day after their issue date; for short-clock forms announced by a portal e-mail, captured within 4 hours of that e-mail.

These can only be met in use: the measures run from the first morning the autopilot is switched on.

Next

- Phase 3 remainder: Liability Register Part II, Rule 86A, DRC-01B/01C Part-B status, bulk filing status, refund folders in the bundle — each after its endpoint is confirmed from the portal's own scripts.
- Phase 4 — Reply Factory I (read and compute): document intelligence on notice PDFs, evidence recipes from the firm's own data, reply rules. Needs the partner's decision on AI processing and the reply positions.